

L2 Strain Gages

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Signed: _____ *K.S.* _____ *R.O.* _____
_____ *Amr* _____ *J.N.* _____

Abstract

Strain Gauges are what is referred to when one attaches an electrical conductor to a material with the desire to measure the changes in electrical resistance in that material as its subjected forces, compressive or tensile. Often used for their portability and accuracy when measuring rapidly changing parameters but often held back by their fragile nature and requiring additional technology to interpret the changes in electrical resistance. This experiment seeks to show how one would install one, then obtain relevant values from it. A digital multimeter will be used to interpret these changes in electrical resistances given by the $120\ \Omega$ strain gauge connected to an aluminum and honeycomb beam with a CFRP skin (woven carbon fiber), using a banana plug connector to connect the strain gauge to the digital multimeter. Installing the strain gauge on a clean Aluminum 6061-T6511 beam, and using an already prepped honeycomb beam with a preinstalled strain gauge, we found that while subjected to tensile forces, the strain gauge showed an increase in resistance of around $0.08\ \Omega$ and $0.1\ \Omega$ for Aluminum and the honeycomb respectively. This same pattern repeating for both beams but in the opposite direction, decreasing by $0.08\ \Omega$ and $0.1\ \Omega$ for Aluminum and honeycomb. This gives us a Young's Modulus of 84 GPa for the aluminum and 74 GPa for the honeycomb. Comparing these to the recorded values, we find a percent error of around 17.9% for the aluminum beam. Plotting the stress vs. strain of both compression and tension, we see that they closely match one another which follows logically since it's the same force just in the opposite direction and makes sense mathematically considering that the change in resistance between compression and tension is the same except for the sign. These results match with available data and show the practicality and accuracy of using strain gauges for similar circumstances that could occur outside of a lab.

Introduction

Two of the fundamental mechanical properties of a material are stress and strain, connected through Hooke's law (1). Together, these properties can tell an engineer how a material will react to compressive and tensile forces. Stress defines the forces that occur within an object, while strain is the ratio between the resulting elongation and the original length (2). Both are also defined in terms of the direction that the outside force is facing compared to the surface; if the force is perpendicular to the surface, they are normal, and if the force is parallel to the surface, they are shear. This paper and the preceding lab focus on collecting strain and then calculating the stress.

$$\sigma = E\varepsilon \quad (1)$$

$$\varepsilon = \frac{\Delta l}{l_0} \quad (2)$$

where:

σ – stress

E – Modulus of Elasticity

ε – strain

Δl = change in length

l_0 – original length

Measuring strain is typically done using one of two device types: an extensometer or a strain gage. Extensometers are typically used for longer specimens, providing both normal and shear strain, but are less precise than strain gages. Strain gages typically provide a strain reading on a much smaller scale, allowing for a more localized measurement, and can be placed on any surface of a specimen. Strain gages are usually composed of an electrical conductor bonded to the surface of the specimen. The resistance in the conductor is then measured as the specimen bends; this resistance, along with a set gage factor, allows strain to be calculated through Equation 3. The strain measured through a strain gage is a more accurate measurement than directly measuring the deflection, but strain gages have some disadvantages. Strain gages are usually more expensive than their mechanical counterparts, are fragile, require supporting equipment, and are difficult to install.

$$\varepsilon = \frac{\Delta R/R}{GF} \quad (3)$$

where:

ΔR – change in resistance

R – resistance

GF – gage factor

Strain gages can be placed on many different surfaces or types of materials. In the testing for this paper, two beams were used. These beams were made from 6061-T6511 aluminum and a honeycomb beam composed of Nomex and CFRP skins. Beams are non-circular members that carry their load perpendicular to their major axis. Both of these beams have a different internal geometry, causing their reactions to forces to be different, resulting in different calculations of stress. While stress can be calculated using Hook's Law, the Modulus of Elasticity is unknown; thus, Equation 4 is used for a conventional beam, while Equation 5 is for a sandwich beam.

$$\sigma = \frac{My}{I} \quad (4)$$

$$\sigma = \frac{M}{ht_fb} \quad (5)$$

where:

M – bending moment

y – distance from neutral axis

I – second moment of area

h – distance between facing skin centers

t_f – facing skin thickness

b – beam width

In this lab, the strain of two beams was measured by installing a strain gage on one end and then measuring the strain through three sets of weights attached to the opposite end. Along with the strain, the stress was calculated using measurements of the beam and the applied force. With both stress and strain, the Modulus of Elasticity was found for both beams through Hook's law, and this was then compared to the accepted modulus.

Equipment/Apparatus

- 6061-T6511 Aluminum strip beam
- Nomex honeycomb beam with CFRP skin
- Vishay instruction bulletins (B-127-14, TT-603, and TT-609)
- 120 Ω strain gages
- Bondable terminals
- Strain gage tape
- 26 AWG strain gage wire
- Banana plug
- Jeweler screwdriver
- Tweezers
- Ruler
- Right triangle
- Caliper
- Degreaser
- Sandpaper (320 grit, 400 grit)
- M-Prep Conditioner A
- Gauze sponge
- Cotton tip applicator
- M-Prep Neutralizer 5A
- M-Bond 200 catalyst
- M-Bond 200 adhesive
- Soldering iron
- Solder flux
- Solvent
- Acetone
- Magnifying lamp
- Gloves
- Goggles
- Scale
- Paper wipes
- Wire strippers
- Scissors
- Scraper knife
- Precision knife
- Digital multimeter
- Mass with hook (1 Kg)

- 2 Additional masses (1Kg each)
- C-clamp



Fig 1: Masses

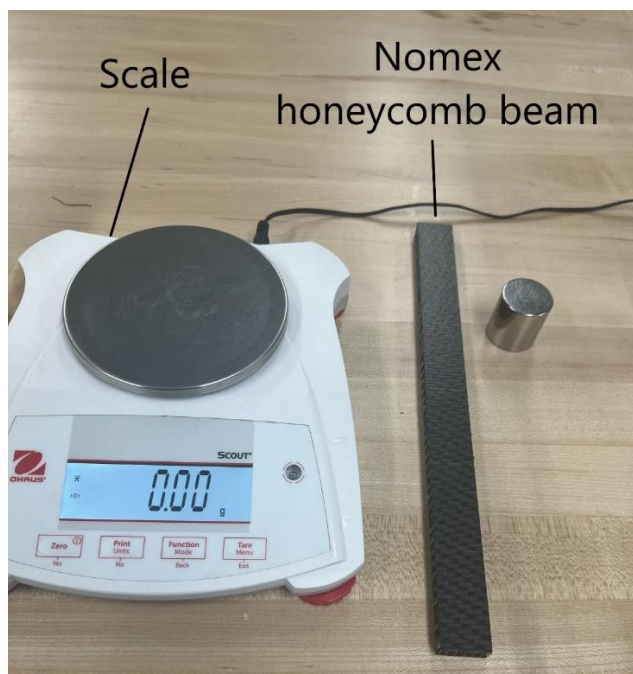


Fig 2: Honeycomb and scale



Fig 3: Tape, gloves, and wipes

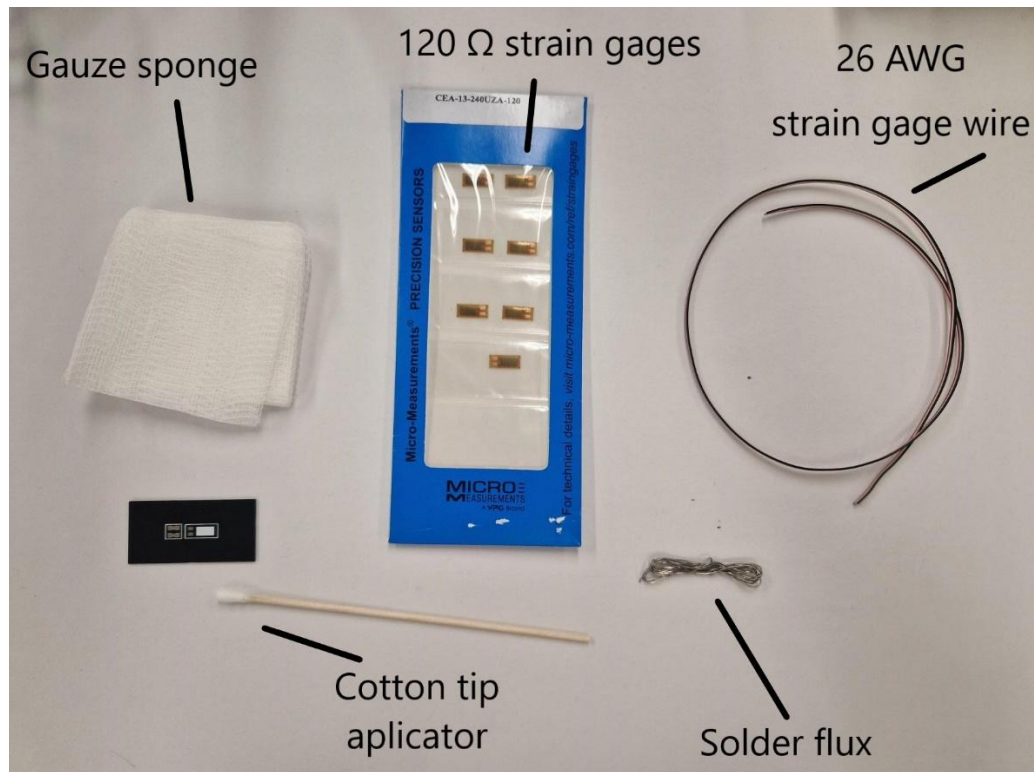


Fig 4(a): Sponge, Strain gages, Wire, Applicator, Flux

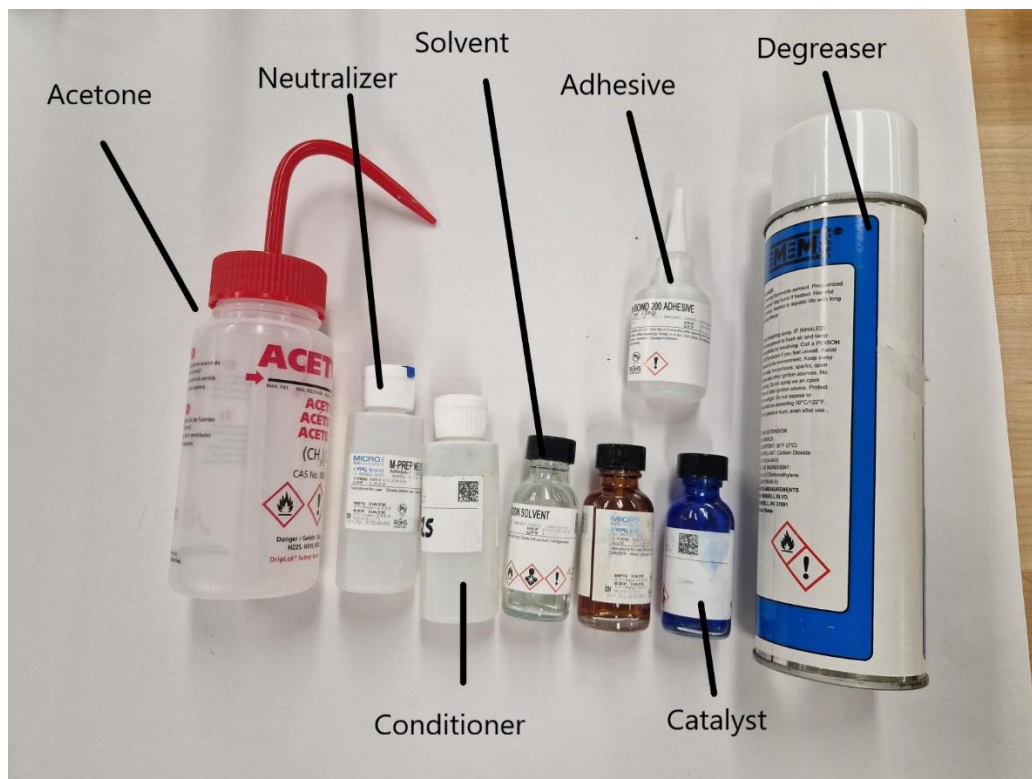


Fig 4(b): Acetone, Neutralizer, Solvent, Adhesive, Degreaser, Conditioner, Catalyst



Fig 5(a): Tools



Fig 5(b): Magnifier lamp

Method

Our group started by measuring the dimensions of the provided aluminum and honeycomb beams, their length, width, and thickness, five times, and taking the average to be the “true” value. The same process was applied when finding the mass of the aluminum and honeycomb beams. The honeycomb beam being an unaltered version as shown in Figure 7.



Figure 6: Mass of Aluminum Bar, Trial 1



Figure 7: Mass of Honeycomb Beam, Trial 1

We then soldered on connecting wires to the strain gauge that would later be used to connect to the digital multimeter after practicing on some test samples using a finished honeycomb beam as an example. Afterwards, using Vishay's Instructions as a guide, we installed the finished strain gauge to the aluminum beam, measuring out around 205 mm from the edge with hole drilled into it.

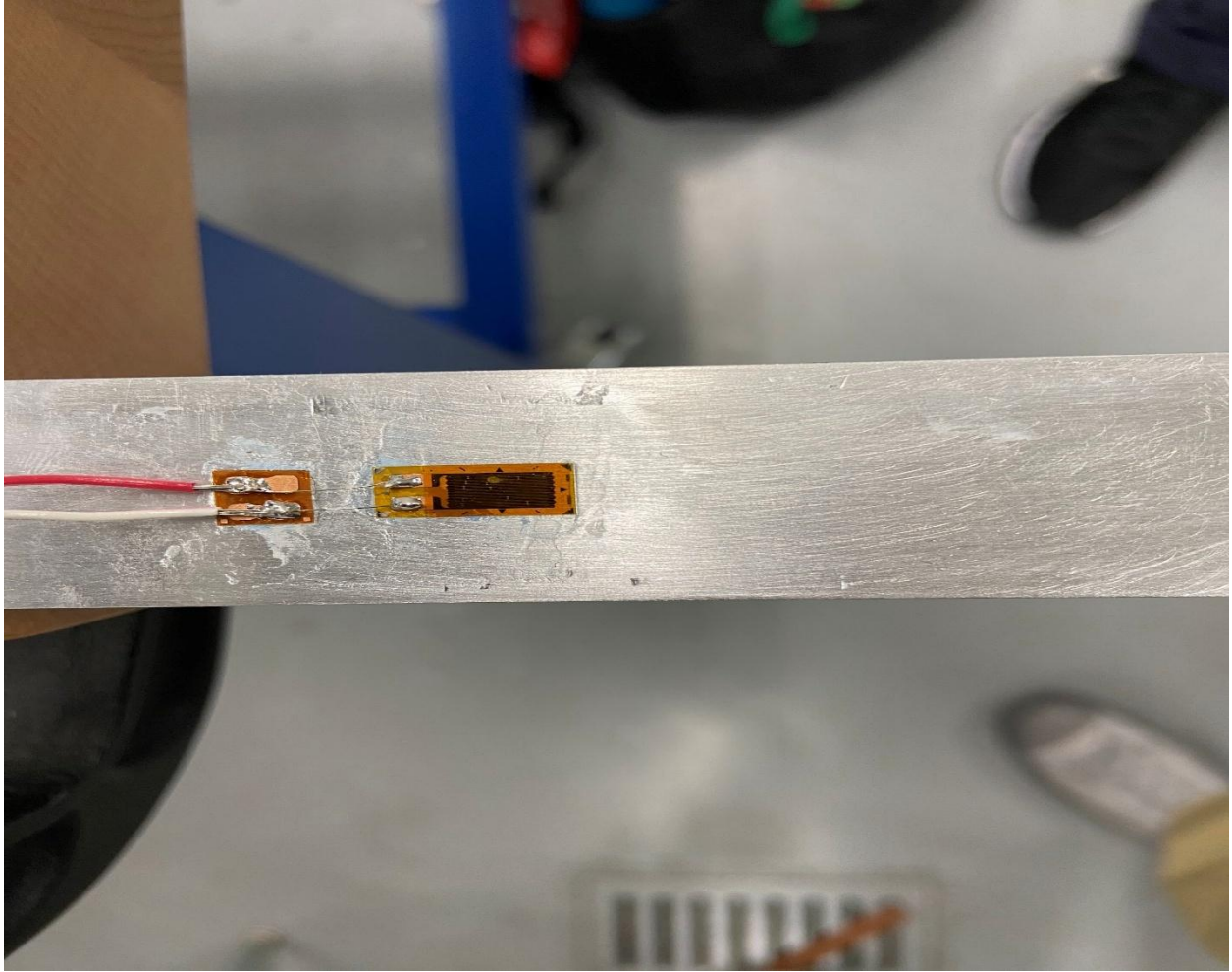


Figure 8: Installed Strain Gauge on Aluminum Beam

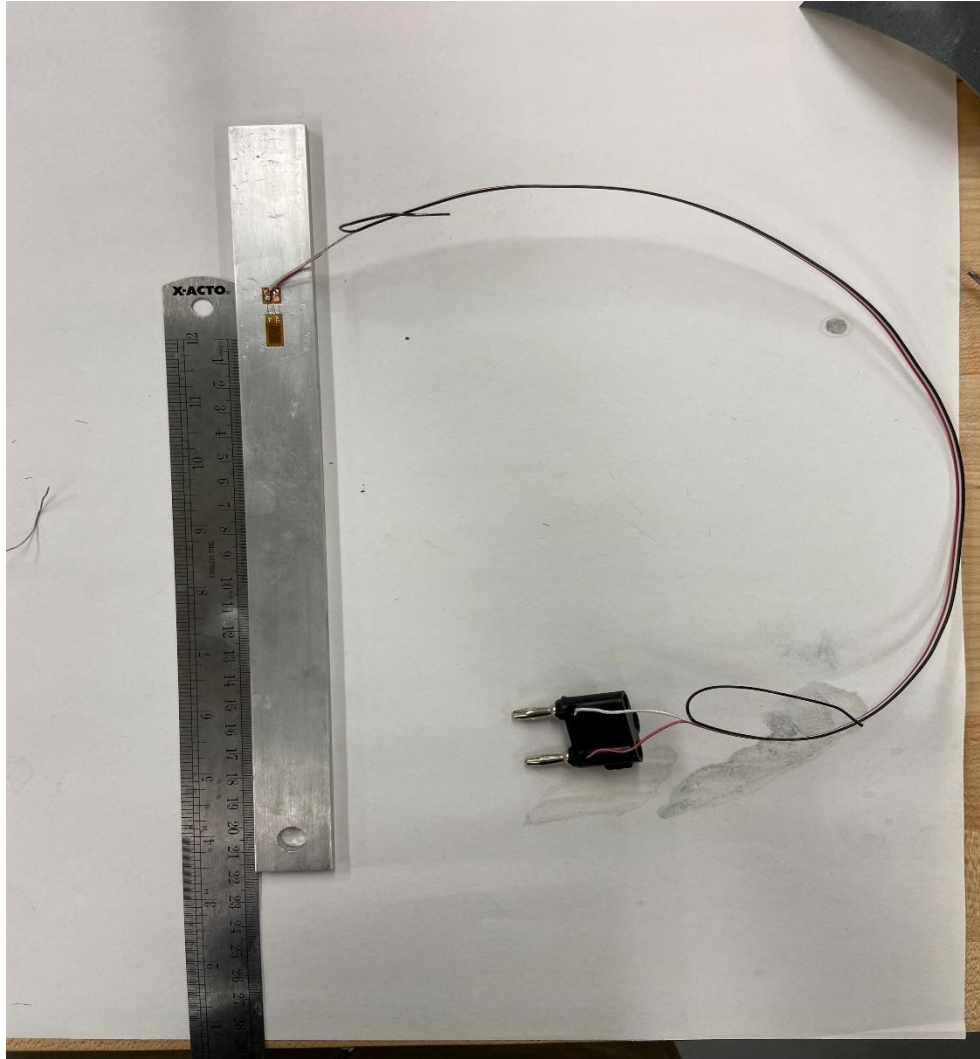


Figure 9: Measuring Distance for Strain Gauge Installation

After confirming that the strain gauge was connected to the bar, we attached the wires to a banana plug connector, which would plug into our digital multimeter, as shown in Figure 9.

After this, we connected the plug to the digital multimeter, making sure the strain gauge was facing up, before clamping the beam to the table with a C-clamp, creating a sort of “diving board”. Measuring the resting resistance as our tensile resistance.



Figure 10: Aluminum Bar, Resting Tension Resistance

Then, proceeding to add weight increasing by one kilogram up to a maximum of three kilograms.



Figure 11: Aluminum Beam in Tension at Max Mass

After collecting all measurements, we removed the weight, unclamped the beam and flipped it upside down, and clamped it again. Measuring this new resistance which would be the resting resistance while in compression before more mass, measuring the new resistance after each additional kilogram worth of mass maxing out at three kilograms.

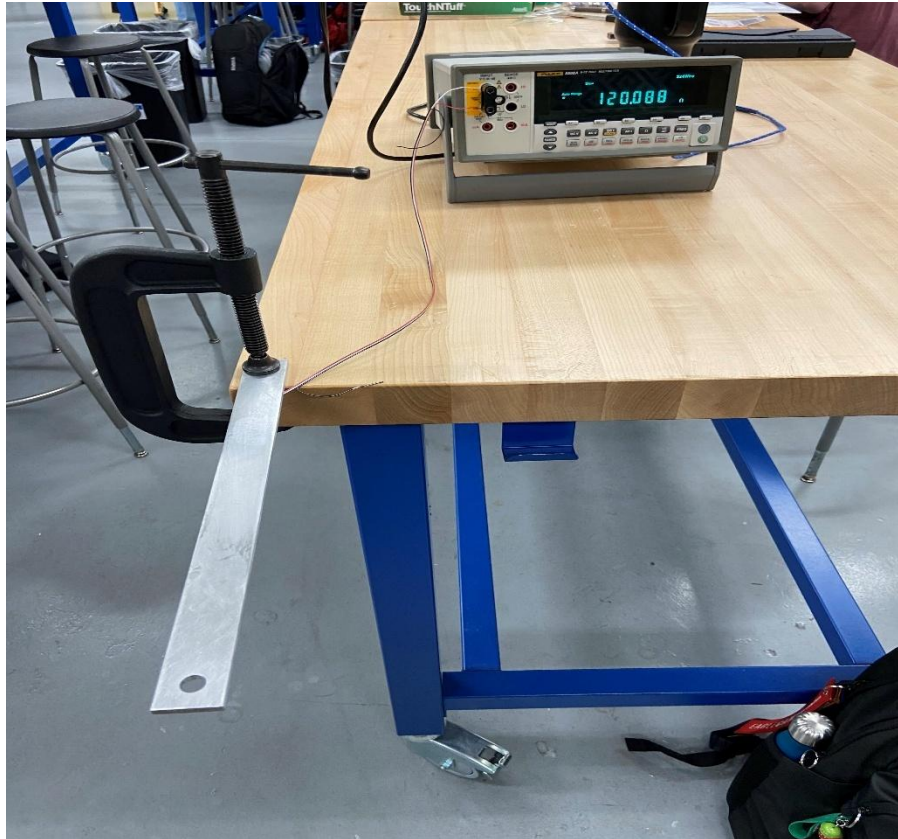


Figure 12: Aluminum Beam with Resting Compression Resistance

After the aluminum beam it was replaced with the honeycomb all the steps were repeated. First it was set in tension with no weight to get the resting tension resistance. Next the weights were placed on one at a time to get the weighted tension resistances. The weight was then removed and the honeycomb was flipped over to measure for the compression resistance. After the non-weighted compression resistance was taken the weighted compression resistance was taken with the three weights. After all the tests the bonded terminal was removed from the aluminum beam using the scraping knife.

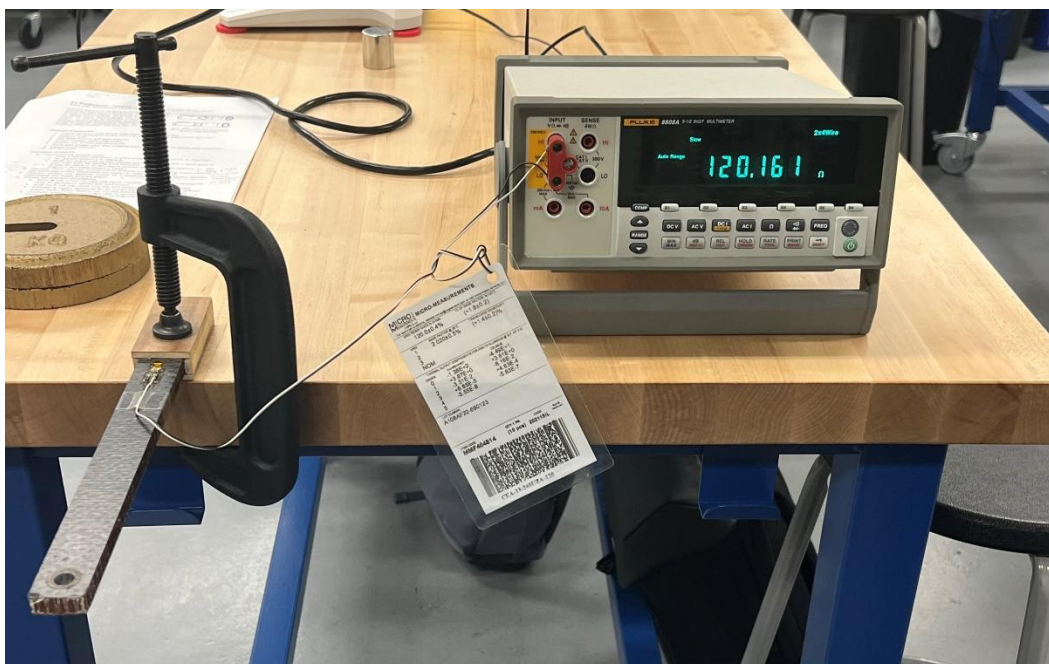


Figure 13: Honeycomb beam with resting tension.

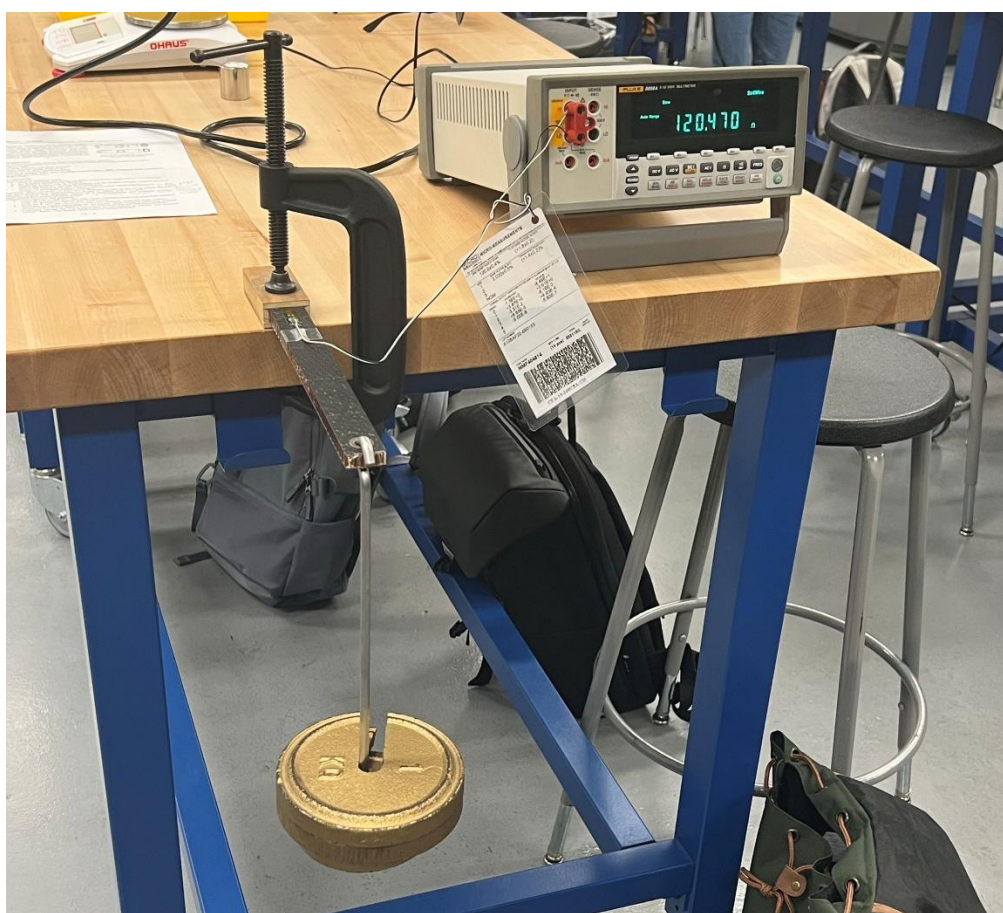


Figure 14: Honeycomb beam with weighted tension



Figure 15: Honeycomb beam with resting compression



Figure 16: Honeycomb beam with weighted compression

Results

Table 1 and 2 shows the dimensions of the aluminum and honeycomb beams our team measured in millimeters over the course of five trials, taking the average of these five trials as the “true” value.

Aluminum Dimensions			
Trial	Width (mm)	Length (mm)	Thickness (mm)
1	25.44	29.9	4.8
2	25.65	29.9	4.97
3	25.51	29.95	4.78
4	25.46	29.9	4.77
5	25.52	29.9	4.8
Mean	25.516	29.91	4.824

Table 1: Aluminum Beam Dimensions

Honeycomb Dimensions			
Trial	Width (mm)	Length (mm)	Thickness (mm)
1	24.5	29.4	10.27
2	24.38	29.4	10.22
3	24.51	29.4	10.26
4	24.49	29.4	10.25
5	24.16	29.4	10.29
Mean	24.408	29.4	10.258

Table 2: Honeycomb Beam Dimensions

Table 3 shows the thickness of the small sample of the honeycomb beam’s skin that has had its core and any adhesive inside removed, measured in millimeters over five trials, the average being used later in our calculations.

<i>CFRP Facing Skin Thickness</i>	
<i>Trial</i>	<i>Thickness (mm)</i>
<i>1</i>	<i>0.34</i>
<i>2</i>	<i>0.33</i>
<i>3</i>	<i>0.35</i>
<i>4</i>	<i>0.32</i>
<i>5</i>	<i>0.34</i>
<i>Mean</i>	<i>0.336</i>

Table 3: Honeycomb Sample Facing Skin Thickness

Table 4 and 5 shows the resistance experienced by the two beams when adding mass to the end of the beams, increasing incrementally by one kilogram, maxing out at three kilograms. This was done twice, once causing the strain gauge to experience compression forces, then another to cause tensile forces.

Aluminum Hanging Mass		
Hanging Resistance (Kg)	Compression Resistance (Ω)	Tension Resistance (Ω)
0	120.088	120.095
1	120.004	120.18
2	119.92	120.263
3	119.837	120.347

Table 4: Aluminum Beam Resistance Test

Honeycomb Hanging Mass		
Hanging Resistance (Kg)	Compression Resistance (Ω)	Tension Resistance (Ω)
0	120.204	120.161
1	120.097	120.264
2	119.993	120.368
3	119.89	120.47

Table 5: Honeycomb Beam Resistance Test

Analysis

The goal of this analysis was to evaluate strain readings from strain gages mounted on two different beams, a Nomex honeycomb core beam with carbon fiber reinforced polymer (CFRP) skins and an aluminum alloy beam (6061-T6511). Resistance variations in the 120 Ω strain gages were recorded for both tension and compression orientations as the experiment applied bending loads by hanging masses at the free end of a cantilever.

Strain can be calculated using the following equation:

$$\varepsilon = \frac{\Delta l}{l_0} = \frac{\frac{\Delta R}{R}}{GF}$$

- Δl – change in length (mm)
- l_0 – original length (mm)
- ΔR – resistance change (Ω)
- R – resistance (Ω)
- GF – gage factor

Sample Calculation (Aluminum beam, Tension, 1 kg load):

$$\varepsilon = \frac{\frac{120.180 - 120.095}{120.095}}{2} = 3.54 \times 10^{-4}$$

(Values are obtained from table 4 in the Results section)

This process was repeated for all loadings and both beams.

Strain Calculations:

Data for the following equations was obtained from tables 4 and 5. Strain was calculated using the following equation:

$$\varepsilon = \frac{\frac{\Delta R}{R}}{GF}$$

Load	Strain - Tension	Strain – Compression
0	0	0
1	0.000354	-0.000350
2	0.000699	-0.000699
3	0.00104	-0.00105

Table 6: Aluminum Beam Strain Calculations

Load	Strain - Tension	Strain – Compression
0	0	0
1	0.000429	-0.000433
2	0.000861	-0.000879
3	0.00129	-0.00131

Table 7: Nomex Honeycomb Core Beam Strain Calculations

Strain vs Load Behavior:

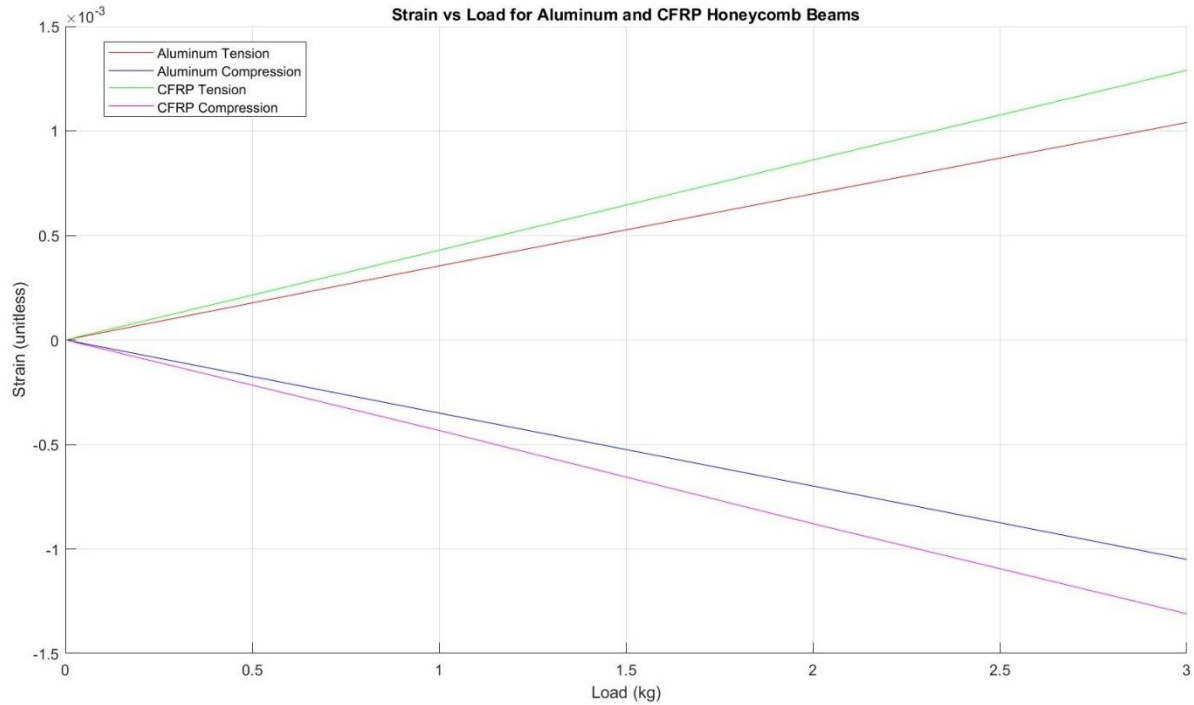


Figure 17: Strain vs Load for different beams

The figure above shows the strain-load relationship for both beams. Both beams showed approximately linear trends in the elastic region. The honeycomb core beam showed reduced strain for the same applied load, indicating its higher stiffness-to-weight performance. On the other hand, the aluminum beam generated nearly symmetric tension and compression responses. Since skin thickness and anisotropy vary in composite materials, the CFRP results also displayed a little bit greater scatter.

Stress Calculations:

Axial stress in a conventional beam can be calculated from:

$$\sigma = \frac{My}{I}$$

- M – bending moment (N·mm)

- y – distance from the neutral axis (mm)
- I – second moment of area about the neutral axis (mm^4)

Elastic facing skin stress for sandwich structure beams can be obtained by:

$$\sigma_f = \varepsilon_f E_f = \frac{M}{ht_f b}$$

- ε – facing skin strain
- E_f – facing skin Young's modulus
- $h = t_f + t_c$ – distance between facing skin centers (mm)
- t_f – facing skin thickness (mm)
- t_c – core thickness (mm)
- b – beam width (mm)

Sample Calculation (Aluminum beam, Tension, 1 kg load):

$$\sigma = \frac{My}{I}$$

$$F = 1 \text{ kg} \times 9.81 \frac{m}{s^2} = 9.81 \text{ N}$$

$$M = F \times L = 9.81 \text{ N} \times 0.299 \text{ m} = 2.93 \text{ N} \cdot \text{m}$$

$$y = \frac{h}{2} = \frac{0.00481}{2} = 0.002405 \text{ m}$$

$$I = \frac{bh^3}{12} = \frac{0.02552 \times 0.00481^3}{12} = 2.36 \times 10^{-10} \text{ m}^4$$

$$\sigma = \frac{2.93 \times 0.002405}{2.36 \times 10^{-10}} = 29732700 \text{ Pa}$$

$$E = \frac{\sigma}{\varepsilon} = \frac{29732700 \text{ Pa}}{3.54 \times 10^{-4}} = 8.39 \times 10^{10} = 84 \text{ GPa}$$

(Dimensions can be obtained from table 1 and 2 in the results section)

Sample Calculation (Honeycomb core beam, Tension, 1 kg load):

$$\sigma = \varepsilon_f E_f = \frac{M}{ht_f b}$$

$$F = 1 \text{ kg} \times 9.81 \frac{m}{s^2} = 9.81 \text{ N}$$

$$M = F \times L = 9.81 \text{ N} \times 0.294 \text{ m} = 2.914 \text{ N} \cdot \text{m}$$

$$h = t_f + t_c = 0.336 + 10.34 = 10.676 \text{ mm} = 0.0107 \text{ m}$$

$$\sigma = \frac{2.914}{0.0107 \times 0.336 \times 10^{-3} \times 24.41 \times 10^{-3}} = 33204635 \text{ Pa}$$

$$E = \frac{\sigma}{\varepsilon} = \frac{33204635 \text{ Pa}}{4.29 \times 10^{-4}} = 7.74 \times 10^{10} = 74 \text{ GPa}$$

Stress vs Strain Behavior:

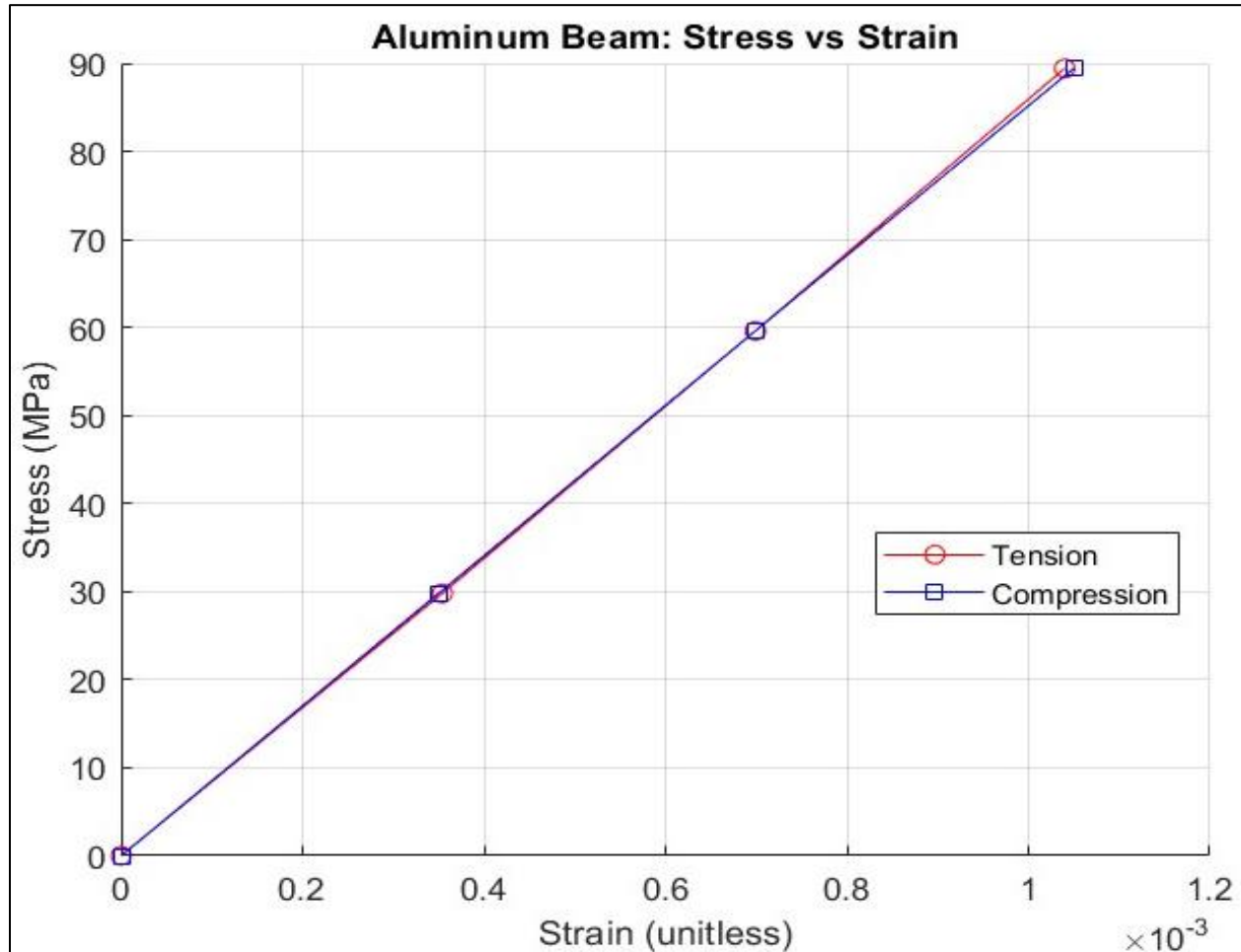


Figure 18: Stress vs Strain Curve – Aluminum Beam

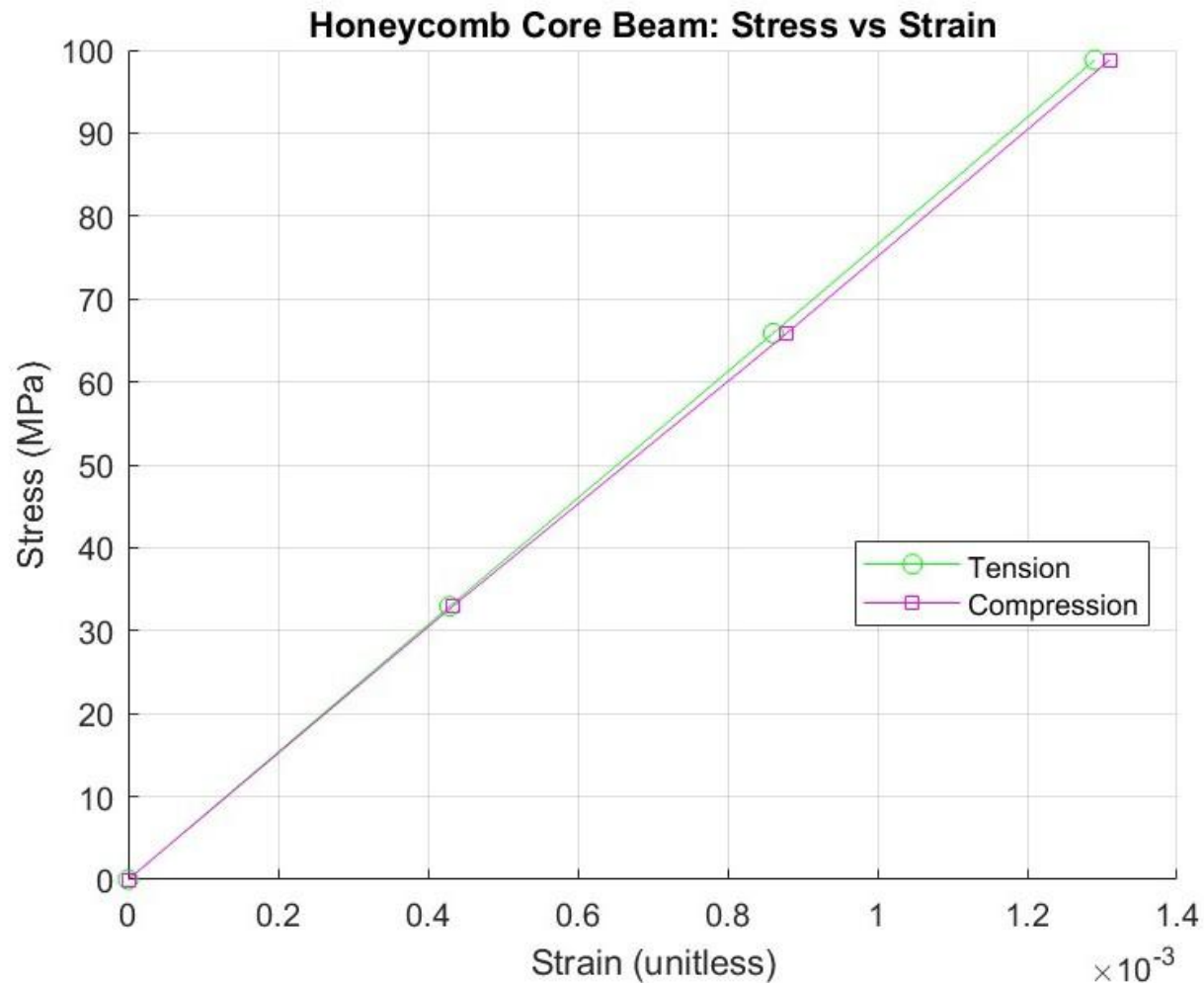


Figure 19: Stress vs Strain Curve – Honeycomb Core Beam

The elastic region of the stress-strain curves showed the predicted linear behavior for both beams. With nearly symmetric tension and compression responses, the aluminum beam demonstrated proper gage installation and consistent material behavior. At the same load, the honeycomb core beam showed less strain, indicating a higher effective stiffness than aluminum.

The calculated Young's Modulus for the Aluminum beam (84 GPa) was closely comparable to the handbook value for 6061-T651 (69 GPa). An approximate 17.9 % difference was recorded between them. For woven carbon fiber, the calculated Young's Modulus (74 GPa) fell between the expected 70 and 120 GPa range.

Sample Calculation (Aluminum beam, Error Percentage):

$$\%Error = \frac{|Measured - Actual|}{Actual} \times 100 = \frac{|69 - 84|}{84} \times 100 = 17.9 \%$$

Some possible sources of error:

- Measurement irregularities could be introduced by the slight misalignment of the gage axis
- Surface preparation could introduce measurement errors
- Due to CFRP's anisotropic nature and possible flaws in the core-to-skin bonding, the compression face is more prone to non-uniform strain

Overall, the findings show the fundamental trade-off in aerospace, i.e., composites offer better stiffness-to-weight with greater sensitivity to installation and manufacturing, whereas aluminum is reliable and simple to test.

Conclusion

This experiment effectively illustrated the use of strain gauges to quantify stress and strain in a Nomex honeycomb composite beam and a typical aluminum beam under bending. Nearly symmetric tension and compression responses from the aluminum beam demonstrated correct gage placement and reliable material characteristics. In contrast, the honeycomb beam demonstrated less strain for the same applied load, demonstrating its higher stiffness-to-weight ratio. However, because of its anisotropic construction and core-to-skin bonding, it also displayed greater variability. The composite Young's modulus was within the predicted range for CFRP materials, aluminum's computed modulus was 84 GPa, which is near to the guidebook value of 69 GPa. In conclusion, the results validated the accuracy of strain gages and provided insight into the trade-offs between metals and composites in structural applications.

Acknowledgements

This experiment was conducted as part of the AE-417 course at Embry-Riddle Aeronautical University. We would like to thank our lab instructor, David Sypeck and teaching assistant Reshma Kubsad for providing the instructions and guidance for this experiment. Each team member contributed to the following sections of the lab report:

A. Bharadwaj: Discussion/Analysis
J. Nguyen Jr.: Method and Results, Analysis
R. Orr: Introduction and Equipment
K. Satvaldy: Conclusion

References

D. J. Syneck, *AE 416/417 Aerospace Structures and Instrumentation Laboratory Bulletins and Supplemental Information*, Embry-Riddle Aeronautical University, Daytona Beach, FL (2024).

Hodgkinson, J. (2012). Testing the strength and stiffness of polymer matrix composites. In *Failure Mechanisms in Polymer Matrix Composites: Criteria, testing and industrial applications* (pp. 129–182). essay, Woodhead Publishing.

Holmes, Douglas. “Mechanics of Materials: Strain.” *Mechanics of Slender Structures RSS*, www.bu.edu/moss/mechanics-of-materials-strain.

Appendix

The appendix consists of the MATLAB code used to generate graphs for the analysis section. The PDF of the code is attached at the end of the document.

```
% Lab 2: Strain vs Load
% Author: Anuranan Bharadwaj

clc;
clear;
close all;

load = [0 1 2 3];

% Strain data
al_strain_T = [0 0.000354 0.000699 0.00104]; % Aluminum tension
al_strain_C = [0 -0.000350 -0.000699 -0.00105]; % Aluminum compression

hc_strain_T = [0 0.000429 0.000861 0.00129]; % Honeycomb CFRP tension
hc_strain_C = [0 -0.000433 -0.000445 -0.00131]; % Honeycomb CFRP compression

% Plotting
figure;
hold on;

plot(load, al_strain_T, '-r');
plot(load, al_strain_C, '-b');
plot(load, hc_strain_T, '-g');
plot(load, hc_strain_C, '-m');

xlabel('Load (kg)');
ylabel('Strain (unitless)');
title('Strain vs Load for Aluminum and CFRP Honeycomb Beams');
legend('Aluminum Tension','Aluminum Compression', 'CFRP Tension','CFRP Compression', 'Location','best');
grid on;
hold off;
```

```
% Lab 2: Strain vs Load
% Author: Anuranan Bharadwaj

clc;
clear;
close all;

% Constants
g = 9.81;

% Loads (kg)
m = [0 1 2 3];
F = m * g;

% Distances clamp to load (m)
L_al = 0.299;
L_cf = 0.294;

% Geometry - Aluminum beam
b_al = 25.52e-3;
h_al = 4.81e-3;
I_al = b_al * h_al^3 / 12;
y_al = h_al / 2;

% Geometry - CFRP sandwich
b_cf = 24.41e-3;
t_core = 10.34e-3;
t_face = 0.336e-3;
h_f = t_core + t_face;

% Strain data
al_strain_t = [0 0.000354 0.000699 0.00104];
al_strain_c = [0 -0.000350 -0.000699 -0.00105];

cf_strain_t = [0 0.000429 0.000861 0.00129];
cf_strain_c = [0 -0.000433 -0.000445 -0.00131];

% Moments
M_al = F * L_al;
M_cf = F * L_cf;

% Aluminum stresses
sigma_al = M_al ./ y_al ./ I_al; % Pa

% CFRP stresses
sigma_cf = M_cf ./ (b_cf * h_f * t_face); % Pa

% Plot Aluminum Stress-Strain
figure;
```

```
hold on;

plot(al_strain_t, sigma_al/1e6, 'o-r');
plot(abs(al_strain_c), sigma_al/1e6, 's-b');

xlabel('Strain (unitless)');
ylabel('Stress (MPa)');
title('Aluminum Beam: Stress vs Strain');
legend('Tension', 'Compression', 'Location', 'best');
grid on;

% Plot Honeycomb Strain ===
figure;
hold on;

plot(cf_strain_t, sigma_cf/1e6, 'o-g');
plot(abs(cf_strain_c), sigma_cf/1e6, 's-m');

xlabel('Strain (unitless)');
ylabel('Stress (MPa)');
title('Honeycomb Core Beam: Stress vs Strain');
legend('Tension', 'Compression', 'Location', 'best');
grid on;
```